Type MS · Describe a control or explain why a step is in a method

Answer 1 of 10 · Question 8 part b.iii · 3 marks

2023 JUN · Paper 01 (regional R) · extract 1 of 2 · find part b.iii in this mark scheme extract

8	(a)	(i)	A = flagellum; B = cell wall/capsule; C = nucleic acid/genetic material/bacterial chromosome/(loop of) DNA;		3 (grad)
		(ii)	pathogen;		1 (cler)
	(b)	(i)	• produced by microorganism/bacteria/fungi; • kills bacteria;		2 (grad)
		(ii)	A *no mark for giving X* • because larger clear zone; • antibiotic diffused from filter paper; • bacteria have been killed;		3 (exp)
		(iii)	• filter paper without antibiotic; • bacterial growth unaffected/no clear zones; • if antibiotic is causing effect;		3 (exp)

Total 12 marks

Type MS · Describe a control or explain why a step is in a method

Answer 1 of 10 · Question 8 part b.iii · 3 marks

2023 JUN · Paper 01 (regional R) · extract 2 of 2 · find part b.iii in this mark scheme extract

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Type MS · Describe a control or explain why a step is in a method

Answer 2 of 10 · Question 9 part b.ii · 1 mark

2019 JUN · Paper 02 · extract 1 of 2 · find part b.ii in this mark scheme extract

9 (a) (i)	• reference to platelets; • fibrinogen converted to fibrin; • (fibrin) mesh/crosslinks/fibres formed;		3
(ii)	• reduced blood flow to heart/cardiac muscle / tissue/cells; • less oxygen/glucose delivered; • less (aerobic) respiration/energy released (by cardiac tissue);		3
(b)(i)	size of blood clot/volume of water /aspirin solution used/temperature/shape of blood clot/width of straw;	Allow valid alternatives	1
(ii)	use water without aspirin dissolved;		1
(iii)	• the more tablets dissolved the more solution collected; • the more tablets dissolved the quicker the solution is collected; • aspirin dissolves/acts on blood clot; • size of blood clot reduced more quickly with more tablets/blood clot dissolves/breaks down faster with more tablets;		Max 3
(c)	Any four from: • aspirin binds to enzyme/aspirin competes (with substrate) for active site; • active site of enzyme changed/blocked; • less/no substrate binds (to enzyme); • reaction involving release of chemicals reduced/stopped; • platelets do not stick together;		Max 4

Total question 9 = 15 marks

Type MS · Describe a control or explain why a step is in a method

Answer 2 of 10 · Question 9 part b.ii · 1 mark

2019 JUN · Paper 02 · extract 2 of 2 · find part b.ii in this mark scheme extract

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Type MS · Describe a control or explain why a step is in a method

Answer 3 of 10 · Question 2 part b · 2 marks

2020 NOV · Paper 02 · extract 1 of 2 · find part b in this mark scheme extract

2 (a) (i)	C; (lipase)		1
(ii)	A; (carbon, hydrogen, oxygen)		1
(iii)	C; (liver)		1
(b)	• show effect of enzyme (1) • show that bile salts on their own don't break down fat (1)	Ignore control	2
(c)	• create alkaline/optimum pH (1) • for enzyme to work quickly (1)	Allow increases/adjusts pH	2
(d)	• increase rate of digestion (1) • with no bile salts change occurred after 15 minutes/digested more slowly in tube A (1) • occurred after 5 minutes with bile salts present/digested more quickly in tube B (1)		3

Total 10 marks

Type MS · Describe a control or explain why a step is in a method

Answer 3 of 10 · Question 2 part b · 2 marks

2020 NOV · Paper 02 · extract 2 of 2 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Describe a control or explain why a step is in a method

Answer 4 of 10 · Question 1 part b.ii · 1 mark

2021 NOV · Paper 01 · extract 1 of 2 · find part b.ii in this mark scheme extract

number				
1	(a) (i)	<table><tr><th>Conditions</th><th>Result/no/scale of bubbles</th></tr><tr><td>Room temperature</td><td>5</td></tr><tr><td>Alkali added</td><td>3</td></tr><tr><td>Acid added</td><td>2</td></tr><tr><td>3°C</td><td>1</td></tr><tr><td>70°C</td><td>0</td></tr><tr><td>no hydrogen peroxide added</td><td>0</td></tr></table> • two column table; • headings; • correct order of results (high to low/ORA);		3 marks
	(ii)	• no repeats/only done once;		1 mark
	(iii)	• results based on judgement/subjective/qualitative;		1 mark
	(iv)	• repeats; • measure volume of oxygen; • using gas syringe/other suitable apparatus;		3 marks
	(b) (i)	untreated/room temperature;		1 mark
	(ii)	control;		1 mark
	(iii)	• reduced/low enzyme activity; • not at optimum pH; • active site changed; • fewer enzyme substrate complexes/substrate won't fit;		4 marks

Total 14 marks

Type MS · Describe a control or explain why a step is in a method

Answer 4 of 10 · Question 1 part b.ii · 1 mark

2021 NOV · Paper 01 · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Describe a control or explain why a step is in a method

Answer 5 of 10 · Question 2 part a.i · 2 marks

2022 JUN · Paper 02 (regional R) · extract 1 of 2 · find part a.i in this mark scheme extract

2	(a) (i)	• a tube containing inhibitor and substrate; • but no enzyme;		2
	(ii)	any three from • concentration of enzyme; • concentration of inhibitor; • temperature; • pH; • same filter in colorimeter;		3
	(b) (i)	axes labelled with units; axes correct way round; plots;		3
	(ii)	suitable curve;		1
	(iii)	similar shaped curve; reaching maximum sooner;		2
	(c)	same maximum never reached; despite increase in substrate concentration;		2

Total 13 marks

Type MS · Describe a control or explain why a step is in a method

Answer 5 of 10 · Question 2 part a.i · 2 marks

2022 JUN · Paper 02 (regional R) · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Describe a control or explain why a step is in a method

Answer 6 of 10 · Question 6 part d · 1 mark

2022 JUN · Paper 02 · extract 1 of 2 · find part d in this mark scheme extract

6 (a)	(type of) antibacterial cleaner;		1
(b)	Any two from the following • use sterilised equipment/named equipment; • heat the inoculating loop (in a Bunsen flame); • do not open the lid of the Petri dish too much; • seal the lid of the Petri dish; (with tape)	Named equipment must be hardware	2
(c)	• to prevent growth/reproduction; • of pathogen/harmful microorganism;		2
(d)	it is a control;		1
			Total 6

Type MS · Describe a control or explain why a step is in a method

Answer 6 of 10 · Question 6 part d · 1 mark

2022 JUN · Paper 02 · extract 2 of 2 · find part d in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Describe a control or explain why a step is in a method

Answer 7 of 10 · Question 7 part d · 3 marks

2024 NOV · Paper 02 · extract 1 of 2 · find part d in this mark scheme extract

7 (a)	• body temperature; • optimum/enzyme is not denatured;	Accept alimentary for MP1	2
(b)	• effect of pH/to see if pH affects enzyme activity; • range of pH used/different pH used/ pH is the independent variable;	***Effect of different pH on enzyme activity = 2 marks***	2
(c)	Any four from: Answers about validity(MP1-4) • The same temperature; • The same volume of enzyme/suspension; • not (valid) because different time used; • the same enzyme is used; Answers about accuracy(MP 5-7) • as difficult to judge endpoints/ when it goes clear; • students' judgement of endpoint may vary/ subjective/ not standardised/ did not use a colorimeter; • timing is too short to reach endpoint/ cannot tell it will go clear if left for longer;	They do not have to mention validity/accuracy to score marks. Must have at least one mark from each section to score 4. Maximum of 3 if answers are from one section	4
(d)	• tube with no enzyme/ denatured enzyme/boiled enzyme; • same volume of boiled enzyme/water /5cm^3 of water/boiled enzyme; • same volume of suspension/same overall volume/ same volume of substrate (in test tube);		3
(e)	• optimum pH; • fastest between 2 and 4;		2
(f)	• stomach; • acidic/low pH/pH2-4/pH2;		2

Total 15 marks

Type MS · Describe a control or explain why a step is in a method

Answer 7 of 10 · Question 7 part d · 3 marks

2024 NOV · Paper 02 · extract 2 of 2 · find part d in this mark scheme extract

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Type MS · Describe a control or explain why a step is in a method

Answer 8 of 10 · Question 5 part c.ii · 2 marks

2019 JUN · Paper 02 (regional R) · extract 1 of 3 · find part c.ii in this mark scheme extract

5(a)	movement of a substance against a concentration gradient/low to high concentration; using energy/ATP;		**(2)**

Question Number	Answer	Additional guidance	Mark
5(b)	• cut pieces of potato to same size/length/mass; • weigh each piece; • place into different solutions; • leave for at least 30 minutes; • remove and dry; • re-weigh/re-measure length; • determine which concentration has no / least change in mass/length;		**(5)**

Question Number	Answer	Additional guidance	Mark
5(c)(i)	0.10 × 100; 7.90 1.3;	Ecf allow correct answer to incorrect figures used in calculation Allow 1.26 / 1.27 for final correct answer	**(2)**

Type MS · Describe a control or explain why a step is in a method

Answer 8 of 10 · Question 5 part c.ii · 2 marks

2019 JUN · Paper 02 (regional R) · extract 2 of 3 · find part c.ii in this mark scheme extract

Question Number	Answer	Additional guidance	Mark
5(c)(ii)	initial mass/volume/size varies; greater change in mass/volume if potato is bigger; provides a standardised result; allows comparison;		**(2)**
Question Number	Answer	Additional guidance	Mark
5(c)(iii)	• plot a graph; • concentration vs percentage change; • find (from graph) concentration at which percentage change is zero/lowest change; • this is the concentration of the contents;		**(4)**

Total 15 Marks

Type MS · Describe a control or explain why a step is in a method

Answer 8 of 10 · Question 5 part c.ii · 2 marks

2019 JUN · Paper 02 (regional R) · extract 3 of 3 · find part c.ii in this mark scheme extract

Paper 2: Subject Human Biology

Question Number	Answer	Additional guidance	Mark

Type MS · Describe a control or explain why a step is in a method

Answer 9 of 10 · Question 4 part c.i · 2 marks

2025 JUN · Paper 02 · extract 1 of 2 · find part c.i in this mark scheme extract

4 (a) (i)		<table><tr><th>Conc (sugar) in au</th><th>Percentage change in mass</th></tr><tr><td>0.00</td><td>8.0</td></tr><tr><td>0.25</td><td>4.0</td></tr><tr><td>0.50</td><td>-1.0</td></tr><tr><td>1.00</td><td>-7.0</td></tr><tr><td>1.50</td><td>-10.0</td></tr></table> • two row/column table; • concentration (sugar) in au; • percentage/% change in mass; • 0.0 to 1.50/1.50-0.00;	Ignore units in the body of the table Accept AU for MP2 Ignore % symbol for MP3 mp3 unit not required for percentage in mass	4
	(ii)	0.50 (au);		1
	(iii)	$\frac{104 \times 7}{100}$; = 7.28 g; =7.3;	Allow 4/100 for 1 mark or 0.28	3
(b)		• change in mass is -1% to +4% • 5% range in mass • between 0.25 and 0.5 au; • accept any answer between 0.375- 0.5; ***an answer of o.45 scores =4 marks***	MP 1-4 are separate MPs unless the answer is 0.45	4
(c) (i)		• dry the cylinder/ remove water/liquid; • affects mass/wrong mass/heavier/inaccurate mass/eq;	Accept weight	2
	(ii)	any two from • volume of (sugar) solution; • temperature; • surface area of cylinders; • same type of potato;	Ignore amount accept volume of liquid	2 **Total =16**

Type MS · Describe a control or explain why a step is in a method

Answer 9 of 10 · Question 4 part c.i · 2 marks

2025 JUN · Paper 02 · extract 2 of 2 · find part c.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Describe a control or explain why a step is in a method

Answer 10 of 10 · Question 2 part a.iii · 3 marks

2022 JAN · Paper 01 · extract 1 of 2 · find part a.iii in this mark scheme extract

Number				
2	(a) (i)	• sample placed in test tube; • Benedict's solution added; • heat in a water bath; • blue to brick red if glucose present/stays blue if absent;		4
	(ii)	• sample from tubing placed in test tube/on spotting tile; • iodine solution added; • brown to blue-black if present/stays brown if absent;		3
	(iii)	• remove any starch/glucose; • spilt when putting solution into tubing; • to avoid contamination/doesn't affect result/investigation/ensure results are valid;		3
	(b)	• water in tube, positive for glucose but negative for starch • contents of tubing should test positive for starch and glucose; • Visking tubing has pores/partially permeable; • size(of pores) allows glucose to pass; • starch can't pass;		5

Total : 15 marks

Type MS · Describe a control or explain why a step is in a method

Answer 10 of 10 · Question 2 part a.iii · 3 marks

2022 JAN · Paper 01 · extract 2 of 2 · find part a.iii in this mark scheme extract

Question number	Answer	Notes	Marks